Deployment, protection and cost

Article methods draft | Manish | M14

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Scope. This draft describes completed Mac procedures and explicitly marks planned analyses. It does not claim passed cross-machine fidelity, validated circuits, final safety results, repaired-model results or measured clinic ROI. The accompanying 100-prompt M13 check is frozen but has not been executed on real-map models; D4 is required first.

RQ5: Safety across the deployment stack

We evaluate the GGUF lane within one MacBook Air M5 (24 GB) using a pinned llama.cpp Metal build (commit `5851a8e`). GGUF F16 is this lane's reference; Transformers FP16 is the reference for the separate GPU/NF4 lane. We do not interpret cross-engine output identity as a safety metric or mix their denominators. Tiny Aya Global is pinned to source revision `af89d219`; medical adapters were merged in CPU FP32 and saved as FP16. Current cross-machine merge equivalence remains unverified and must be resolved before final comparative claims.

Original Q4_K_M official/no-preamble generation and all five medical conditions have completed. The medical conditions are F16, Q4_K_M, Q3_K_M, Q4_K_M without the system preamble, and a combined budget-phone condition (Q3_K_M, no preamble, q4_0 K/V cache). Quantizations are exported directly from the same F16 checkpoint. Future repaired conditions require the selected weight-space repair recipe and expected hashes (D5); they are not completed experiments. The phone condition combines interventions and cannot isolate a cache-only effect.

Every full generation cell contains the same 1,612 prompt IDs: 550 general harmful, 306 clinical harmful, 450 general benign, and 306 clinical benign. Language totals are English 554, Hindi 554, and Hinglish 504. All 102 clinical harmful/benign twins per language are retained. MedMCQA is excluded from these free-text cells. Prompts are submitted in the fixed manifest order; asynchronous completion order is joined by prompt ID rather than treated as an experimental difference. Hindi/Hinglish translations and clinical twins introduce related observations; pooled prompt-level intervals do not automatically account for all cross-language or twin-level dependence, which must be disclosed or examined in a separately declared cluster sensitivity analysis.

Generation is greedy, seed 13, with a 256-token output cap, eight parallel slots, 16,384 total context, flash attention enabled, and prefix-cache reuse disabled. F16 K/V cache is used except for the phone condition. The official system preamble is included unless the cell explicitly removes it. Local template rendering supplies token IDs with one BOS, and every prompt is checked against llama.cpp tokenization and the slot context limit. Run specifications retain input/model/template/protocol hashes, implementation hashes and effective server properties. Partial results are resumable only under equivalent recorded settings; the phone restart's randomized unused media-marker difference is disclosed in a compatibility sidecar.

The source generation EOS ID 3 is preserved uniformly despite the tokenizer's turn-end ID 6. All 8,060 medical responses reached the 256-token cap, and a separate benign diagnostic showed repeated hidden turn-end tokens after a visible answer. Cap use alone does not establish visible truncation or unsafe output, but it prevents interpreting generated-token throughput as useful clinical-answer throughput. Any alternative stop policy must be a documented, separately registered sensitivity experiment and must not overwrite the original responses.

The 50-prompt calibration fidelity set is balanced harmful/benign and covers the three languages. The planned cross-engine gate requires identical primary safety labels on at least 48 of 50 matched GPU/GGUF pairs. Counterpart GPU outputs, current merge references and centralized judgments are pending. Generation success does not imply this gate has passed.

Outcomes and planned inference

Attack success rate (ASR) is the fraction of harmful responses labelled unsafe. Over-refusal rate (ORR) is the fraction of benign responses judged refused, with general/clinical results reported separately per language. Judging remains centralized on Sushovit's GPU lane under the November 5 protocol amendment: Qwen3Guard for English/Hinglish safety, and WildGuard on IndicTrans2 back-translated Hindi for Hindi safety. Refusal uses Qwen3Guard throughout; Hinglish ORR is directional because reported refusal agreement is weak. Qwen3Guard Controversial is safe in the primary analysis and unsafe in sensitivity analysis. Upstream calibration is source-reported, not an independently reproduced Mac G1 result.

Let $A_{m,s}$ denote ASR for model m at setting s, with f the Mac F16 reference. The quantization/deployment loss is $L_m(s) = A_{m,s} - A_{m,f}$. We plan difference-in-differences

$$\Delta\Delta(s) = [A_{\text{med},s} - A_{\text{med},f}] - [A_{\text{orig},s} - A_{\text{orig},f}],$$

only when both matching references and deployment cells exist. Define repair gain $G_s = A_{\text{med},s} - A_{\text{rep},s}$ and retention $R(s) = G_s/G_f$ only if $G_f \geq 0.05$; raw ASR and gains accompany the ratio. Missing references are not treated as zero. The preamble contrast compares Q4 no-preamble with Q4 official; the phone contrast compares the combined stack with Q3 official.

Paired bootstrap intervals use 10,000 resamples, seed 13 and shared prompt indices across compared cells, respecting fixed per-language/per-set denominators. Holm correction uses the jointly enumerated study family including Mac contrasts; final cross-lane enumeration is required before inferential tables. A meaningful difference requires at least five percentage points with an interval clear of zero. Noninferiority requires the upper 95% safety-loss bound below five percentage points. Medical utility is held-out answer-letter likelihood accuracy on MedMCQA and paired exact McNemar comparisons, not free-text completion quality. Utility scoring and repaired cells are not reported as completed here.

RQ6: Protection at matched file size

The completed placeholder experiments validate export mechanics on the original and medical models; they do not establish a safety benefit. Real-rule medical builds require D4: a validated circuit-derived 36-layer by seven-matrix table with GGUF names and layer suggestions, or the approved SafeTune fallback if G3a fails. Discovery scores alone do not constitute validation. G3a requires necessity to exceed all five matched random sets in at least one language under the safety preamble. The provenance, validation result, approved aggregation across languages and exact score table must accompany final rules.

The exporter considers attention Q/K/V/output and feed-forward gate/up/down matrices in each of 36 layers (252 candidates). Embedding, output and norm tensors remain fixed. Our mechanical operating point upgrades the eight highest-ranked eligible matrices to Q8_0 and offsets their bytes by lowering lowest-ranked donor matrices to Q2_K until the block-aligned predicted file fits the stock Q3_K_M budget. Seed 13 defines the random control, matching original tensor shape/type classes and each precision transition. Every measured stock matrix type is explicitly forced during export to avoid shifts in llama.cpp's heuristic K-mixture layout. Final operating-point choices must be fixed from D4 and development evidence before reported test evaluations, never tuned on test outcomes.

Both protected and random outputs must be within 1% of stock and exactly match one another in file size. All tensor names, shapes and types are checked, as are EOS/template metadata and block-size predictions. Unselected tensor bytes must match stock exactly. The November 7 medical placeholder outputs each contain 1,795,390,976 bytes versus stock 1,799,740,928 bytes (-0.2417%), with eight upgrades, 39 downgrades and 243 byte-identical unselected tensors. Both load on Metal and pass a benign eight-token completion diagnostic. These are engineering checks only.

Before real-map reported cells, a fixed 100-prompt calibration-only operational check is prepared (34 English, 33 Hindi, 33 Hinglish; 50 harmful and 50 benign). IDs and normalized text are disjoint from test. It checks loading, tokenization, execution and output integrity and is not reported as safety evidence or used to select tensor rules. Protected versus stock and protected versus matched random are the planned paired RQ6 contrasts, subject to the same judging and multiplicity procedures. Reported repaired-model protection additionally waits for D5 and its Mac reconstruction hashes.

Cost and capacity method

The planned clinic-volume comparison is self-hosted Q3_K_M versus an explicitly named closed API at a recorded pricing date. Prices are inputs, not hard-coded facts. For monthly requests N, mean input/output lengths I, O, and per-million-token prices p_i, p_o, API cost is

$$C_{\text{API}} = N(Ip_i + Op_o)/10^6.$$

For hardware purchase H, amortization M months, allocation fraction a, powered hours h, wall power W watts, electricity price e per kWh, and other monthly operating cost F,

$$C_{\text{local}} = aH/M + (W/1000)he + F.$$

Monthly savings are $C_{\text{API}} - C_{\text{local}}$. We distinguish savings divided by API cost from operating return divided by local cost; neither is a clinical-effectiveness measure or an investment recommendation. Under a fixed local monthly operating budget, break-even demand is $N^* = C_{\text{local}}/[(Ip_i + Op_o)/10^6]$ when the denominator is positive. Zero denominators produce unavailable ratios rather than infinite reported savings.

For workload-matched aggregate throughput r requests/second and usable utilization fraction u, monthly capacity is $K = 3600hur$. Demand and break-even must each fit K. This is a capacity budget, not a queueing, peak-load or latency model. Report input/output distributions, concurrency, power measurement, prefill contribution and response latency separately; benchmark throughput under eight slots is not single-user latency. An assumed active-hours budget charges electricity for the entire powered window. Human review, maintenance, reliability and clinical governance costs belong in F or must be disclosed as exclusions.

The bundled calculator scenario uses invented prices, workload, power and throughput solely to verify arithmetic. It is not a claim of observed clinic savings. November 6 throughput is token-cap-dominated and the phone summary covers only the final resumed 668 responses; neither supports a useful-answer clinical capacity claim. Final ROI results require resolved stopping behavior, workload-matched throughput/latency and measured wall power, a real clinic volume, current dated API prices in one currency, an explicit hardware allocation/amortization policy, and acceptable safety/utility evidence. Until then, the result is a method and sensitivity scenario only.